

# **Auto Scaling Groups.**

## **1. What is ASG?**

An **Auto Scaling Group (ASG)** in Amazon Web Services automatically manages a group of Amazon EC2 instances.

It ensures:

- The **right number of instances**
- At the **right time**
- Based on **demand**

## **2. Why ASG is Used**

Without ASG:

- Manual scaling → slow + error-prone
- Over-provision → waste money
- Under-provision → system crash

ASG solves all three:

- High availability
- Cost optimization
- Fault tolerance

To create an ASG we have to create on Launch Template using the template the EC2 Creation will be easier. And ASG uses the Template to Maintain the Desired State.

EC2 > Launch templates > Create launch template

### Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

#### Launch template name and description

Launch template name - *required*

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

#### Template version description

Max 255 chars

**Auto Scaling guidance** [Info](#)  
 Select this if you intend to use this template with EC2 Auto Scaling  
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags  
 ► Source template

#### ▼ Summary

**Software Image (AMI)**  
-

**Virtual server type (instance type)**  
-

**Firewall (security group)**  
-

**Storage (volumes)**  
-

[Cancel](#) [Create launch template](#)

#### Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

#### ▼ Application and OS Images (Amazon Machine Image) - required

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Then I want to create the ASG Using this Template.

### Launch template [Info](#)

[Switch to launch configuration](#)

**Launch template**  
 Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

**Version**  
 [Create a launch template version](#)

|                                        |                                                                                 |                                     |
|----------------------------------------|---------------------------------------------------------------------------------|-------------------------------------|
| <b>Description</b><br>-                | <b>Launch template</b><br><a href="#">mynewtemplate</a><br>lt-089542bb742ec4165 | <b>Instance type</b><br>t3.micro    |
| <b>AMI ID</b><br>ami-0a162bb3d7a786dcc | <b>Security groups</b><br>-                                                     | <b>Request Spot Instances</b><br>No |

Then we have to configure network settings.

### Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

**VPC**  
 Choose the VPC that defines the virtual network for your Auto Scaling group.

[Create a VPC](#)

**Availability Zones and subnets**  
 Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

- aps2-az1 (ap-south-2a) | subnet-0b2632107df8e87bf [X](#)
- aps2-az2 (ap-south-2b) | subnet-06d0954c109f3161d [X](#)
- aps2-az3 (ap-south-2c) | subnet-0ffaddacae2501675 [X](#)

After that we have to select the with Load Balancer or without Load balancer.

**Load balancing** [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

**Select Load balancing options**

☒ **No load balancer**  
 Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ **Attach to an existing load balancer**  
 Choose from your existing load balancers.

☐ **Attach to a new load balancer**  
 Quickly create a basic load balancer to attach to your Auto Scaling group.

**VPC Lattice integration options** [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

**Select VPC Lattice service to attach**

☒ **No VPC Lattice service**  
 VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**  
 Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

**Application Recovery Controller (ARC) zonal shift - new** [Info](#)

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**  
 New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

**Health checks**

Created ASG.

**Auto Scaling groups (1)** [Info](#) Last updated less than a minute ago [Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

| <input type="checkbox"/> | Name  | Launch template/configuration   | Instances | Status | Desired capacity | Min | Max | Availability Zones   |             |
|--------------------------|-------|---------------------------------|-----------|--------|------------------|-----|-----|----------------------|-------------|
| <input type="checkbox"/> | myasg | mynewtemplate   Version Default | 2         | -      | 2                | 1   | 2   | 3 Availability Zones | See details |

After the ASG Creation the instances are automatically started creating using the ASG with the help of Launch Template.

**EC2** > **Instances** Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets: Running [Find Instance by attribute or tag \(case-sensitive\)](#)

[Instance state = running](#) [Clear filters](#)

| <input type="checkbox"/> | Name | Instance ID         | Instance state | Instance type | Status check      | Alarm status                  | Availability Zone | Public IPv4 D |
|--------------------------|------|---------------------|----------------|---------------|-------------------|-------------------------------|-------------------|---------------|
| <input type="checkbox"/> |      | i-03db1718cc32507dc | Running        | t3.micro      | Initializing      | <a href="#">View alarms +</a> | us-east-1b        | ec2-54-211-6f |
| <input type="checkbox"/> |      | i-0e9bf71698e4cc24c | Running        | t3.micro      | 3/3 checks passed | <a href="#">View alarms +</a> | us-east-1d        | ec2-52-55-21f |

It will recreate itself new Two Instances.

**EC2** > **Instances** Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets: Running [Find Instance by attribute or tag \(case-sensitive\)](#)

[Instance state = running](#) [Clear filters](#)

| <input type="checkbox"/> | Name | Instance ID         | Instance state | Instance type | Status check | Alarm status                  | Availability Zone | Public IPv4 D |
|--------------------------|------|---------------------|----------------|---------------|--------------|-------------------------------|-------------------|---------------|
| <input type="checkbox"/> |      | i-0e9bf71698e4cc24c | Running        | t3.micro      | Initializing | <a href="#">View alarms +</a> | us-east-1d        | ec2-52-55-21f |

After we have to set some threshold to create an instance properly After utilizing the CPU of the instance.

Using the Dynamic Scaling Policies we can create the threshold for the ASG.

This is my Dynamic Policy with the 1% of CPU Utilization.

When the 1% Utilization is Hold Up For 5 mins It will Create One New Instance.

The image shows two screenshots from the AWS IAM console. The top screenshot is the 'Create dynamic scaling policy' form. It has the following fields: 'Policy type' set to 'Target tracking scaling', 'Scaling policy name' set to 'Target Tracking Policy', 'Metric type' set to 'Average CPU utilization', 'Target value' set to '1', 'Instance warmup' set to '20 seconds', and a checked checkbox for 'Disable scale in to create only a scale-out policy'. The bottom screenshot shows the 'Dynamic scaling policies (1)' list. It contains one policy named 'Target Tracking Policy' with details: Policy type: Target tracking scaling, Enabled or disabled: Enabled, Execute policy when: As required to maintain Average CPU utilization at 1, Take the action: Add or remove capacity units as required, Instances need: 20 seconds to warm up before including in metric, and Scale in: Disabled.

**Create dynamic scaling policy**

Policy type  
Target tracking scaling

Scaling policy name  
Target Tracking Policy

Metric type [Info](#)  
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.  
Average CPU utilization

Target value  
1

Instance warmup [Info](#)  
20 seconds

☒ Disable scale in to create only a scale-out policy

[Cancel](#) [Create](#)

**Dynamic scaling policies (1)** [Info](#) [Actions](#) [Create dynamic scaling policy](#)

**Target Tracking Policy** ☐

Policy type  
Target tracking scaling

Enabled or disabled  
Enabled

Execute policy when  
As required to maintain Average CPU utilization at 1

Take the action  
Add or remove capacity units as required

Instances need  
20 seconds to warm up before including in metric

Scale in  
Disabled

**Predictive scaling policies (0)** [Info](#) [Actions](#) [Create predictive scaling policy](#)

I have Give Fake Stress To Create The Instance.

```
[root@ip-172-31-4-180 ec2-user]# stress --cpu 1 --timeout 300
stress: info: [28081] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
```

This is the created Instance After the Creation Dynamic Scaling Policy.

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Instances (3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Running

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

| <input type="checkbox"/> | Name | Instance ID         | Instance state | Instance type | Status check      | Alarm status | Availability Zone | Public IPv4 D |
|--------------------------|------|---------------------|----------------|---------------|-------------------|--------------|-------------------|---------------|
| <input type="checkbox"/> |      | i-03db1718cc32507dc | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1b        | ec2-54-211-66 |
| <input type="checkbox"/> |      | i-0e9bf71698e4cc24c | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1d        | ec2-52-55-218 |
| <input type="checkbox"/> |      | i-058924225dc6f11a5 | Running        | t3.micro      | Initializing      | View alarms  | us-east-1f        | ec2-98-82-128 |